

Assignment 1 Database

Group Name: G033

Allocated Tutor Name: Dwi

Group Members:

1. **Rishabh Ray**
2. **Yuxin Chen**

- The relationship between LGA and Branch did not clear states on the case study, we assume LGA may not open a branch.
- A manager will choose a branch as home branch, but manager amount might be less than branch amount, so we assume some branches may do not have a manager choose it as home branch, the branch can be chosen by a manager as home branch while some branch might not be chosen.
- As FAQ mentioned on the ed form, we assume the library might not have all book editions, the library will miss some book editions, so book copy might not contain some book editions.
- We assume all library Branch are opening for a while (not just open for public) and have at least one borrower.
- We are not sure whether borrower need to register before they go inside the library (will be register as borrower before they borrow anything), and we assume those people just go to library and loan nothing are visitors, and borrower will be register when they loan any book from the library, so borrower will only exist when they loan any book.
- A book copy may just be published or just arrived in the library, then nobody already borrowed it, so we assume not all BOOKCOPY will occur in the LOAN, so its minimum is none.

Assumptions for Logical Model:

- Assumed that the phone number of the person will be up to 10 digits.
- Manager ID is kept as a 7-character id
- People may book the same book at the same time, and hence the reserve book system uses date time
- Added surrogate keys to simplify key management and avoid overly complex composite keys.

- Books are marked with a specific status and are excluded from the regular loan cycle.